FIREWATCH 2026 · WSU INNOVATION WORKS × LOCKHEED MARTIN SKUNK WORKS

Firewatch

Detect → Confirm → Respond

Early wildfire warning for rural Kansas volunteer fire districts, built to work on one bar of cell.

COTTONWOOD CREEK · 450 SQ MI · 18 VOLUNTEERS



THE PROBLEM

A fire nobody sees for 31 minutes



31 min
from ignition to the first 911 call

+8 min
satellites see Kansas wildfires a median 8 minutes after someone reports them (2016–2025)

2–3 mi
of fire front when the first truck arrives at 2:52

And a system that cries wolf over March pasture burns gets switched off by April.

Detect → Confirm → Respond

Detect

One event per fire

- Co-op line faults on fire-weather days
- Satellite hotspots: VIIRS, MODIS, GOES, NGFS
- Texted reports and tower smoke sensors

Confirm

No drive-out needed

- ML score tells wildfires from burns and flares
- Two independent signals to escalate
- Nearest neighbors texted: "See smoke? 1 yes, 2 no, 3 my burn"

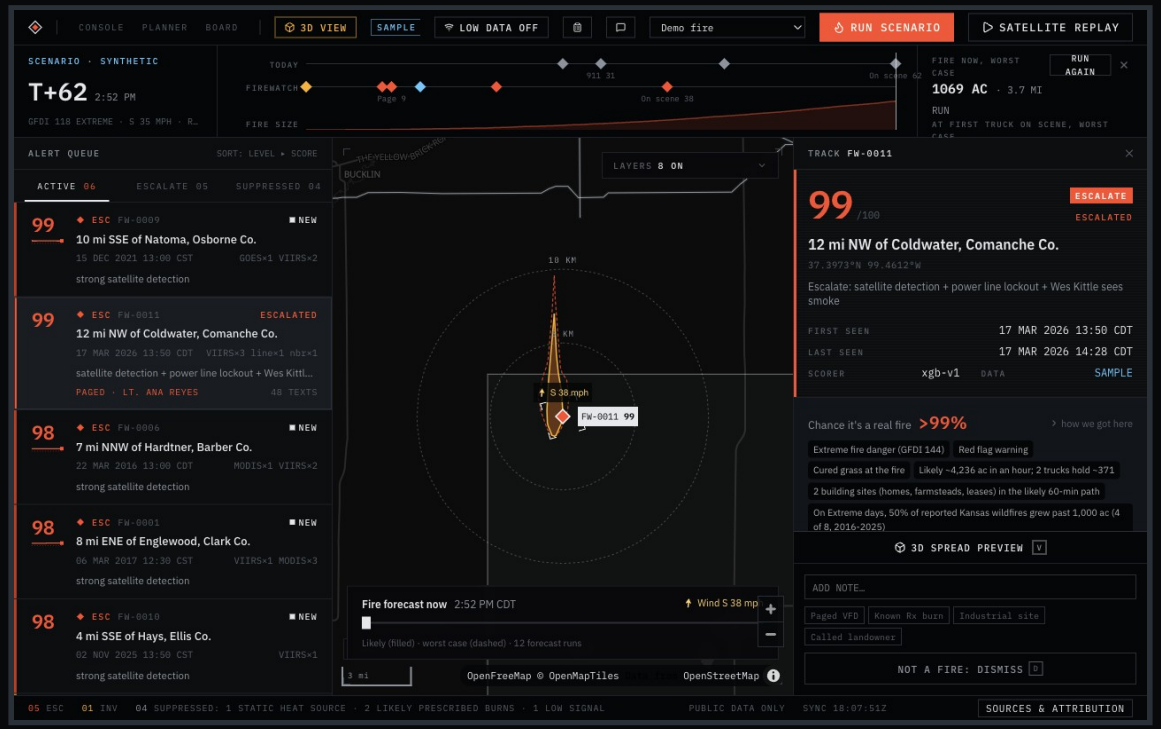
Respond

Before the first engine

- Duty officer replies PAGE by text
- Downwind ranchers get direction and arrival time
- Mutual aid drafted when crews are short

Humans decide every page. Every signal, text and decision is logged on the event.

Escalated in 8 minutes, paged in 9



T+0 Line fault on a red-flag day starts an event; neighbors are texted

T+0 Duty officer gets an FYI

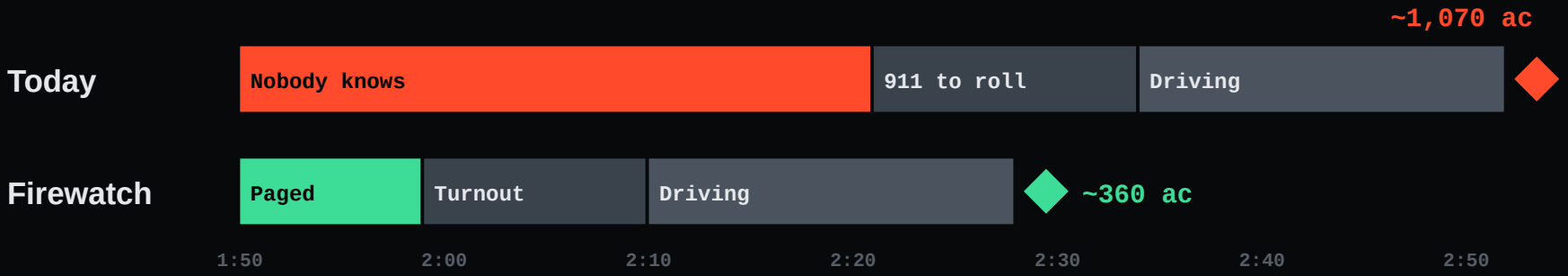
T+8 Satellite confirms: score 99, Escalate

T+9 Duty officer replies PAGE; county dispatch pages the district

T+12 Mutual aid requested; downwind ranchers warned with arrival time

SAME FIRE, SAME CREW

First truck on scene 24 minutes sooner



T+9

district paged

today: 911 call at T+31

T+38

first truck on scene

today: T+62

~360 ac

burned when it arrives

today: ~1,070 acres

Fire sizes are our spread forecast's median for the packet's red-flag Tuesday, not field data. The forecast is back-tested on real Kansas fires (Anderson Creek 2016, Starbuck 2017).

NO CRYING WOLF

Quiet on burns, loud on wildfires



Machine-learning scorer, tested on 2023–2025 fires it never saw

10 years

of Kansas satellite fires, matched to 224 reported wildfires and 490 burns

2 sources

before anyone is paged; one source only sends the duty officer an FYI

0 texts

for a burn the landowner registered, unless it escapes

BUILT FOR ONE BAR OF CELL

Reaches people where cell doesn't

Plain text messages

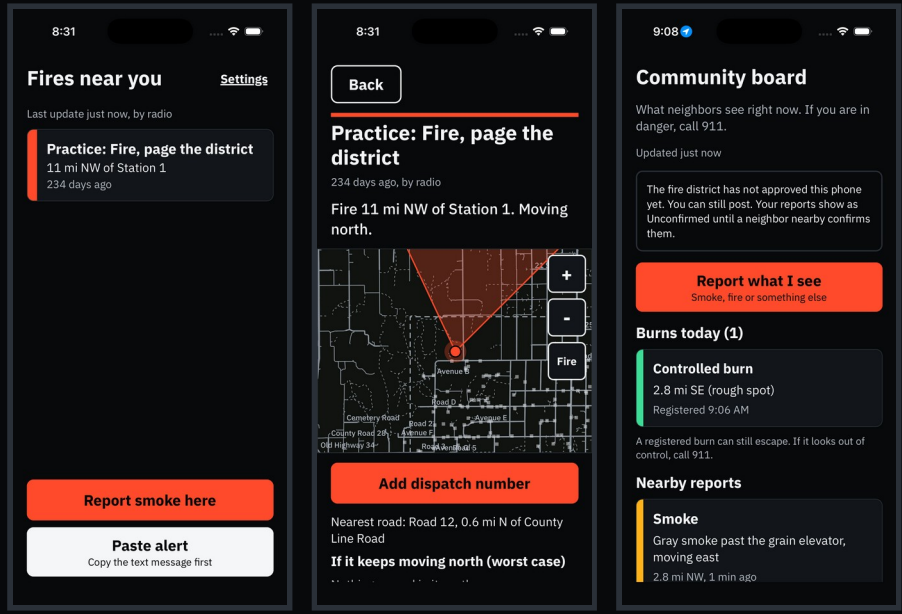
Reply PAGE, 1, 2 or 3. Works on any phone, about 1 cent a text.

LoRa mesh radios

Meshtastic radios carry the same texts to trucks and ranches out of coverage. Tested with real radios.

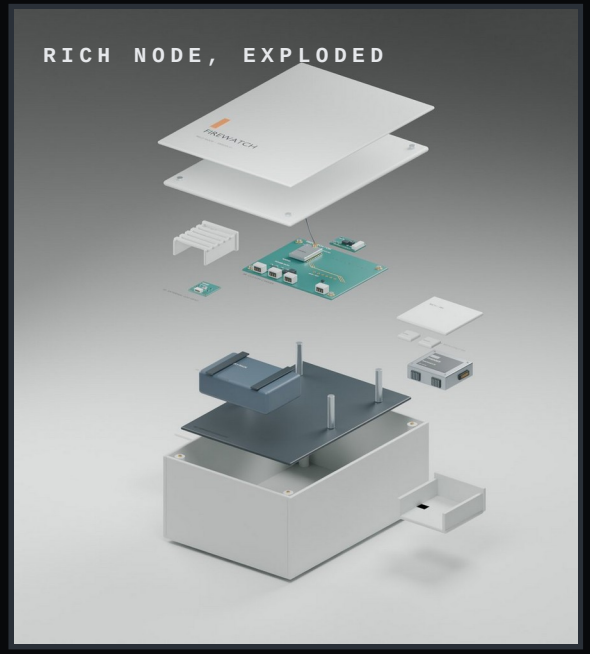
Phone app

Alerts by text or radio, an offline map of where it's heading, and a board for neighbors' reports.



THE SENSOR NODE · CAD REVISION 2

One box, two tiers, on a stake or a pole





Bare · ground stake



Rich · pole band

$136
bare node at ~100 units

$204
rich node, adds smoke sensing

Solar
no wiring, relays the mesh

BARE

Gas, temperature, humidity, pressure (Bosch BME688). 2 W panel, 4,400 mAh.

RICH

Adds a particle sensor that sees smoke (Plantower PMSA003I). 5 W panel, 6,600 mAh.

Blender design model, not a routed board. Detection distance, power budget and radio range are what the field burn trial measures.

Plan sensors for any Kansas county

Places 15-acre sensor plots where fires start and where the wind carries their smoke, from ten years of fires, power lines, tallgrass, terrain and hourly winds. Each plot says why it was chosen.

$15.6k — Cottonwood Creek: 10 plots, 90 nodes, 5 relays, with spares

$136 / $204 — custom-board bare and rich nodes

$0 / month — free public data, no subscriptions



WHERE IT STANDS

Built this weekend, tested on real data

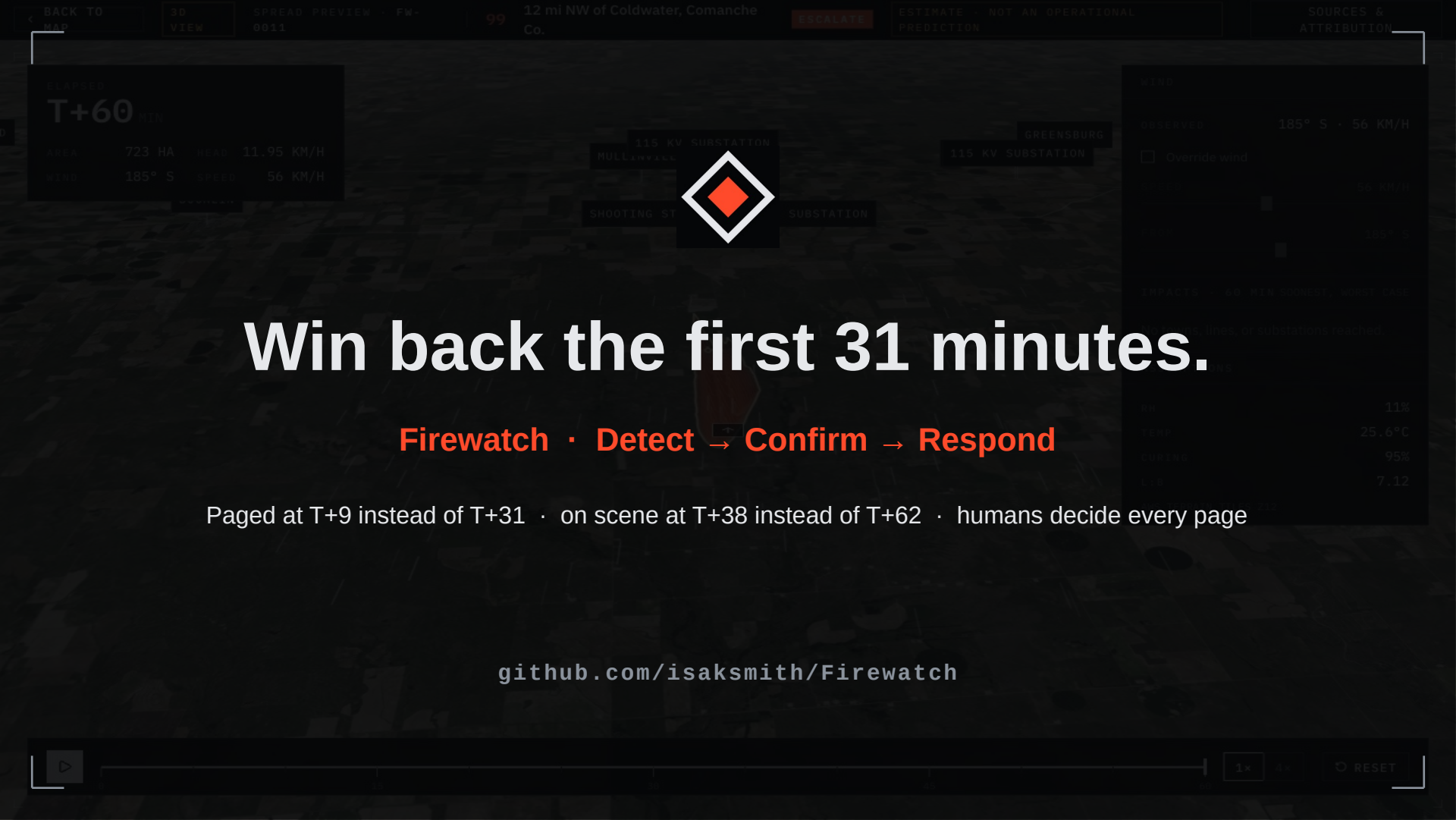
BUILT AND TESTED

- 598 automated tests pass
- Scorer trained on 10 years of Kansas fires
- Spread forecast back-tested on real fires
- 13 live public data feeds checked today
- Packet scenario runs end to end, every text logged

NEXT

- Field burn trial to measure sensor range
- Live texting on real phones
- Pilot with one Kansas district before spring burn season

To run a district
Free public data, about 1 cent a text, optional sensors from $136 a node



Win back the first 31 minutes.

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Paged at T+9 instead of T+31 · on scene at T+38 instead of T+62 · humans decide every page

`github.com/isaksmith/Firewatch`